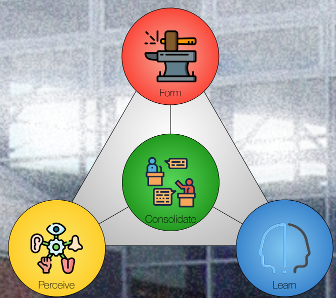
5 SOLUTION

ESSENCE – CHAPTER 8

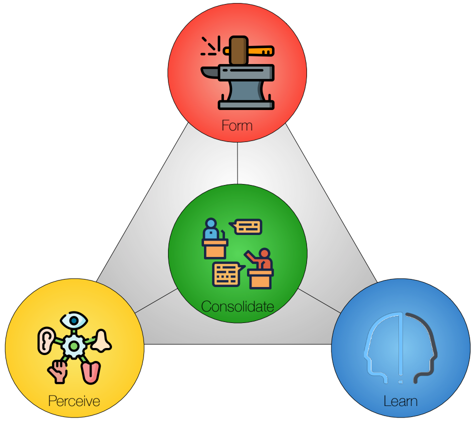
IVAN AAEN
INSTITUT FOR DATALOGI
IVAN@CS.AAU.DK

SWI 2026



Agenda

- Exercise 4 Follow-up.
- Chapter 6: Solution.
    - Resolution.
        - Inquiry.
        - Ends and Ends-in-view.
    - Solution
        - Prospect, warrant, and backing.
    - Solution Scenarios.
- Next time:
    - Exercise 5: Solution Scenarios.
    - Lecture 6: Horizon.

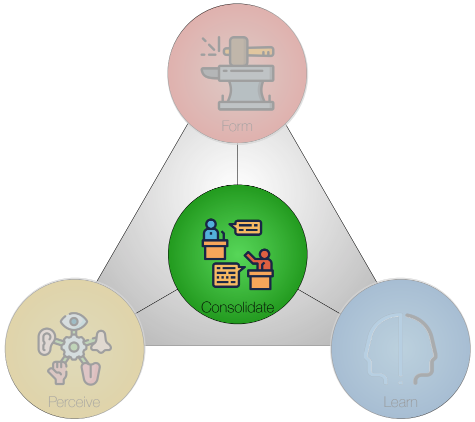


Follow-up: Exercise 4

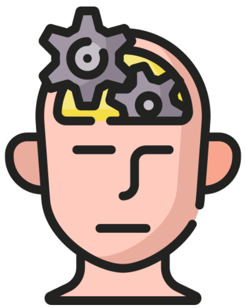
Exercise 4 (Leverage Scenarios)

Use the results from Exercise 3 in this exercise.

- Suggest leverage points for your project. Use the categories in Table 7.1 for inspiration, or develop your own.
- Evaluate and select leverage points. Use PCRT-analysis as shown in Table 7.3 and/or Table 7.4.
- Develop axes for the Leverage Scenarios as described in Section 7.3 (Tables 7.5, 7.6, 7.7, and 7.8).
- Develop Leverage Scenarios as illustrated in Figure 7.1.
- Explain the Leverage Scenarios you find in each quadrant.
- Choose one scenario to work with from now on. You can choose one quadrant or combine several if relevant.
- Discuss the most important leverage points you plan to use in your project. Enter these leverage points in the Leverage cell (Section 3.6).
- Discuss the capabilities obtainable through these leverage points. Enter these capabilities in the Capabilities cell (Section 3.4).
- Discuss the Inner Environment of your design. Enter the key modules in the Inner Environment cell (Section 3.5).



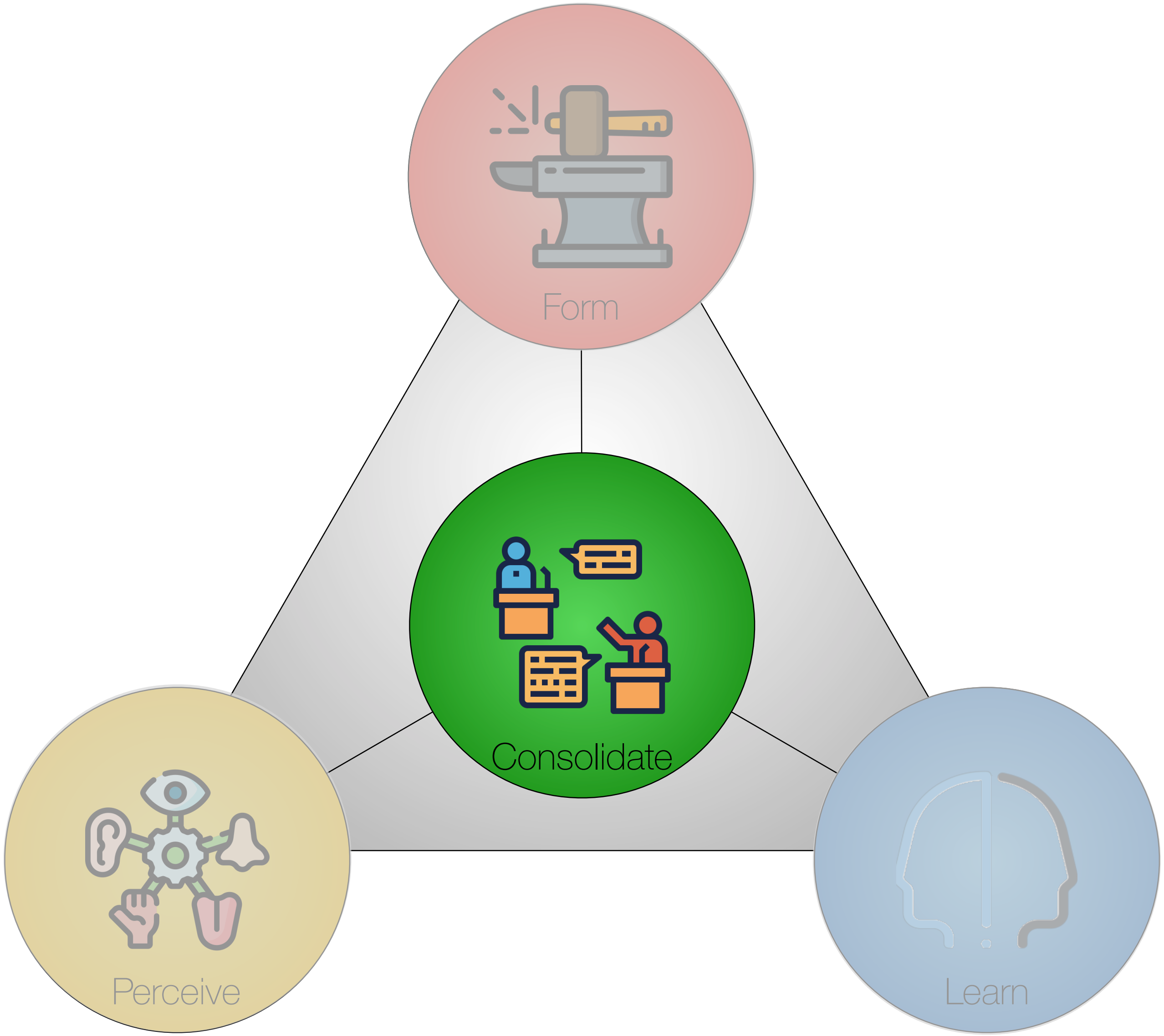
Consolidate



Ideas are anticipated consequences (forecasts) of what will happen when certain operations are executed under and with respect to observed conditions.
John Dewey (1859-1952)

Four Core Activities

- *Perceive* – understanding the problem and its context.
- *Form* – designing the constructive parts of a solution.
- *Consolidate* – integrating the design contributions into a clear whole that addresses the situation.
- *Learn* – appraising the design.

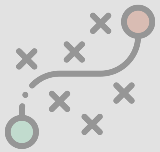


Icons from Flaticon.com

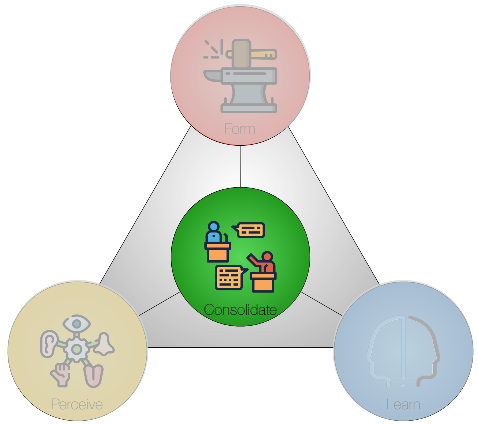
Three Levels of Abstraction

Rationale – the logical basis that makes actions meaningful and helps reason about strategy and tactics for solving the overall problem.

Strategy – the master plan for solving the overall problem, including the scope of the problem, the key components for building the solution, and qualifications regarding limitations and constraints that may affect the utility or acceptability of a solution.



Tactics – actions planned in accordance with the strategy to achieve specific ends.



Deweyan Pragmatism

Solution

The Consolidate column in the Problem-Solution Canvas focuses on how and why we expect our contribution to solve the problematic situation.

Our understanding of the problem and how to solve it develops throughout a project.

This is, therefore, a learning process.

- Dewey refers to this process as *inquiry*.

What we aim for might change in the course of the project.

- The goal we aim for is what Dewey calls *end-in-view*.
- What we end up having is what he calls an *end*.

Inquiry Defined

Inquiry is the controlled or directed transformation of an indeterminate situation into one that is so determinate in its constituent distinctions and relations as to convert the elements of the original situation into a unified whole.

John Dewey (1859-1952): *Logic: The Theory of Inquiry* (1938) p104-105.

Inquiry Characteristics

At its core, inquiry is *transactional*, *open-ended*, and *inherently social*.

It tries things out to see what works, embraces changes in conditions, scope, and focus, and sees the process as a social one in which different insights and appreciations interact.

Inquiry employs two kinds of operations

- One dealing with *ideational* or *conceptual* subject-matter – to make the situation understandable and see ends for resolving it.
- One with activities involving *existential* subject-matter. Observations and experiments that interact with and modify objects and events in a situation.

Ends and Ends-in-View

- An end-in-view is an idea of an end to be reached. It represents the purpose of taking a step.
- The end-in-view is the anticipated consequence of taking a step.
- An inquiry aims to resolve a problematic situation through existential changes.

Ends vs. Means

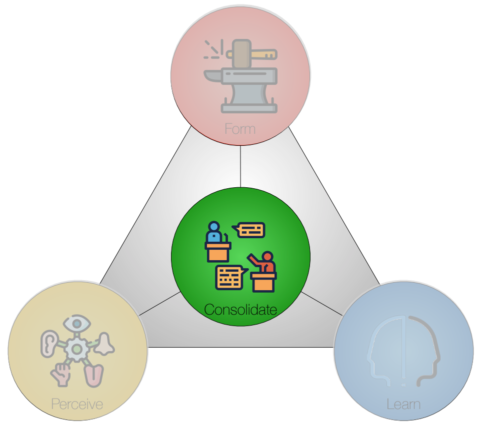
- *"End" is a name for a series of acts taken collectively – like the term army.*
- *"Means" is a name for the same series taken distributively – like this soldier, that officer.*
- *To think of the end signifies to extend and enlarge our view of the act to be performed.*
- *It means to look at the next act in perspective, not permitting it to occupy the entire field of vision.*
- *The end thus re-appears as a series of "what nexts."*

John Dewey (1859-1952): *Logic: The Theory of Inquiry* (1938) p108-109.

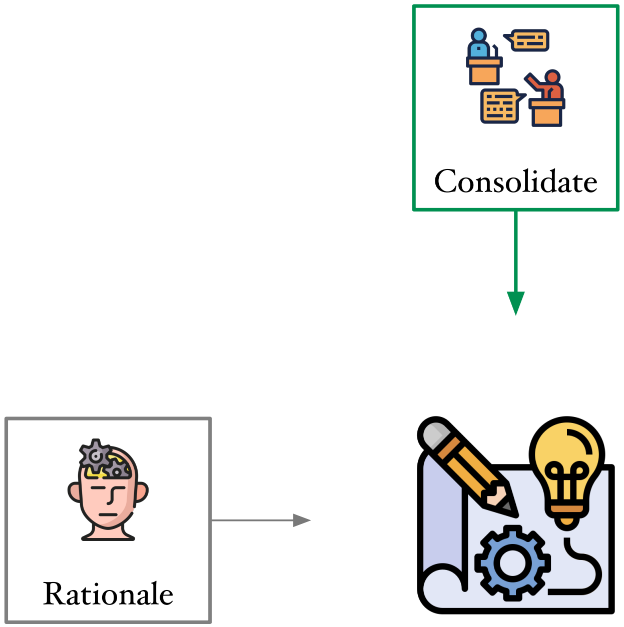
Means and Ends

Material means	Procedural means	Ends, Ends-in-view
• Components built (modules for counting people) • Builds and releases (an experimental prototype) • Platforms and libraries chosen (drivers for sensors) • Objects and events (ticket-holders, concert venue) • Data available (contact data for ticket-holders and staff) • Infrastructure, digital ecosystem (relevant Android and IOS versions) • *Backlog items* (concert manager module) • *Presumed objects and events* (ticket-holders, staff, managers) • *Data presumed* (standardized venue information)	• Acceptance criteria (acceptable accuracy) • Test cases and results (Reject access for ticket-holder) • Prototypes (population density map for venue) • *Sprint (agreed procedure)* • *Planning (agreed procedure)* • *Test- and evaluation procedures (agreed procedure)* • *Daily stand-up meetings (agreed procedure)* • *Refactoring (agreed procedure)* • *Retrospectives (agreed procedure)* • *Demonstrations (agreed procedure)*	• Sprint goal (venue modeling with density thresholds) • Prospect (This app prevents crowding in indoor venues) • Plan (outline of expected iterations in the project) • Requirement (accuracy for population density) • User story (venue manager checks density map) (Ends-in-view may also serve as procedural means)

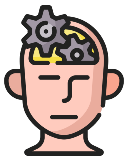
Examples of means and ends



Solution



Consolidate Terminology

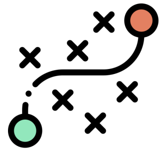


Solution

Why the problem is resolved, why it is important, and why the result is appealing.

Evolvability

The capacity to adapt to outer environments or to add new leverage points to the inner environment.



Merit

The value offered by the capabilities. The limitations of the solution and why these limitations are acceptable.

Prospect

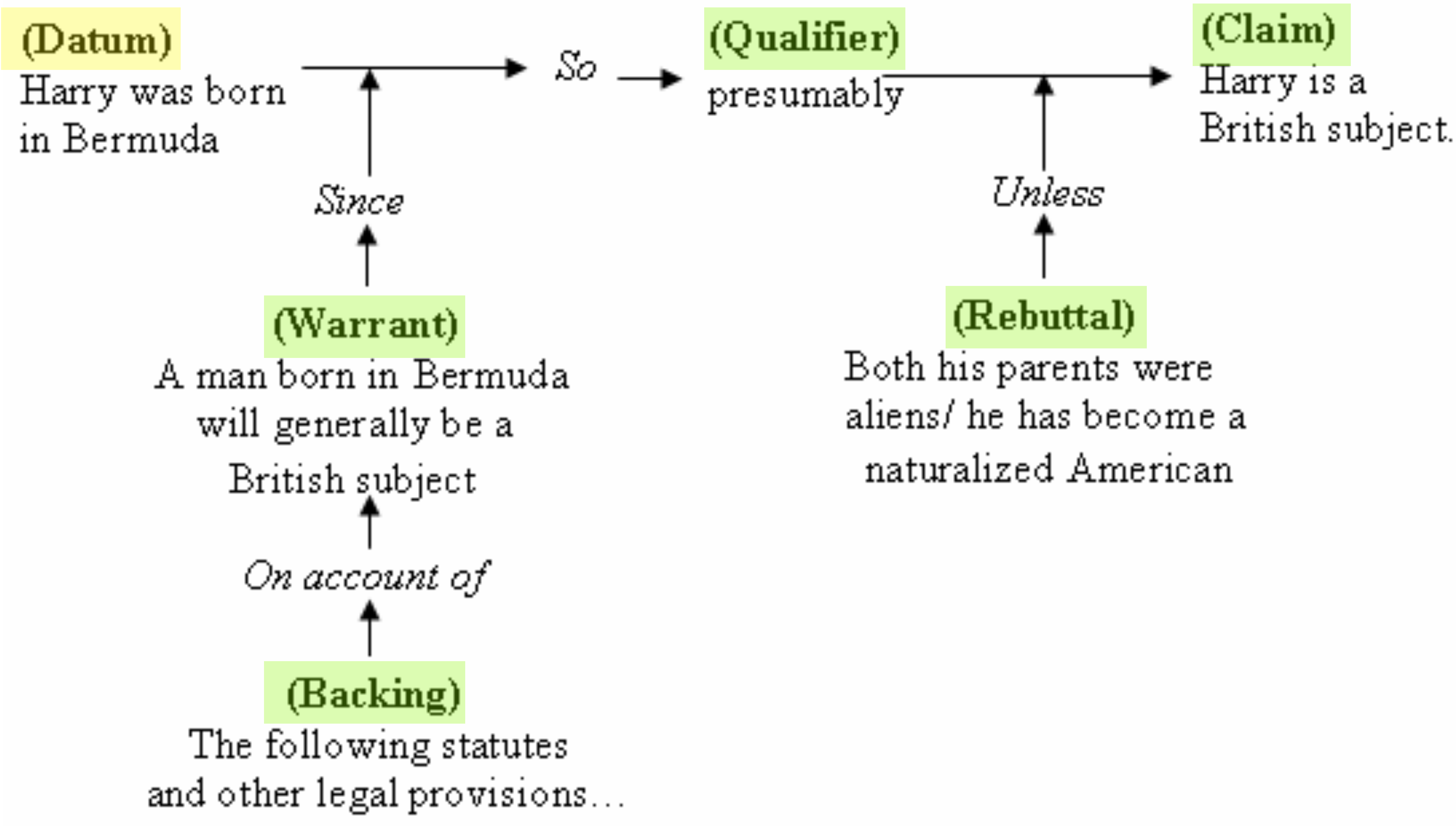
- What we believe to be a solution if the products developed in the project are put to proper use.
  - What Dewey would call our overall end-in-view.
- A prospect reflects an understanding of the problem and an adequate way to solve it.
- May span several iterations or sprints.
- A project may have the same prospect from start to end.

Prospect:
A mental picture of something to come.

Warrant and Backing

- *Warrant* – Why solving the problem is important.
- *Backing* – why the prospect is an attractive and useful solution to the problem.

Toulmin Diagram



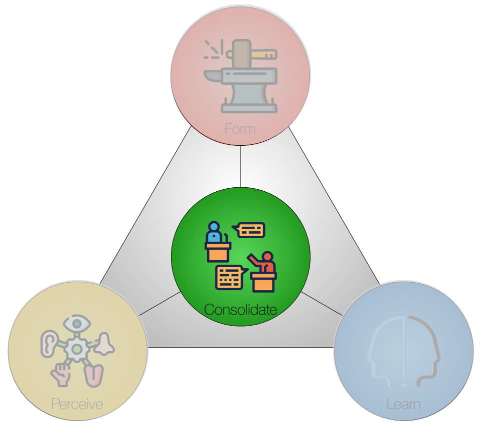
Toulmin, S. E. (2003). *The uses of argument* (Updated ed.). Cambridge, U.K.; New York: Cambridge University Press.

The Toulmin Model of Argumentation in Essence

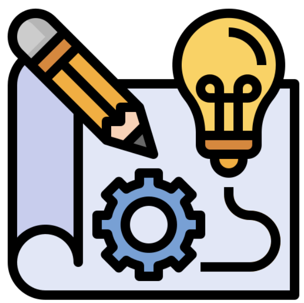
Core Activity	**Perceive**	**Form**	**Consolidate**	**Learn**
Rationale (Why?)	• Problem (Datum)	Leverage	Solution • Prospect (Claim) • Warrant • Backing	Horizon
Strategy (What?)	Outer Environment	Inner Environment	Evolvability • Diffusibility • Adoptability	Properties
Tactics (How?)	Manifestations	Capabilities	Merit • Value • Reservation • Rebuttal	Mission

Illustration: ClusterDetect

	Perceive	Form	Consolidate	Learn
Ratio-nale	**Problem** In crowded areas, safety could be compromised	**Leverage** Technology: AI [DFNet] Components: [Thermal camera] Information: Problem spots Human resources: [On-the-ground staff]	**Solution** Prospect: ClusterDetect reduces congestion and disruption Warrant: Densely populated areas are common and dangerous Backing: The Contribution is effective and inexpensive	**Horizon** The project creates strengths and opportunities - The Problem is understood and generic - The Leverage is future-proof - The Prospect solves the Problem, - The Warrant suggests growth - The Backing suggests motivation to adopt
Stra-tegy	**Outer Environment** External service: [Access control] Implements: Venue information system Repository: Traffic information People: [Informants]	**Inner Environment** Density monitor [Thermal?] Panic and disruption locator General modules for on-the-ground staff and visitors [Customized module for on-the-ground staff]	**Evolvability** Diffusibility: ClusterDetect is highly specialized and has low diffusibility towards new markets Adoptability: ClusterDetect is limited to the current market, but it can address new needs and thereby strengthen existing market positions	**Potential** The design has strategic potential: - Outer Environment is generalized - Inner Environment is near-decomposable - The Evolvability is limited but non-critical
Tactics	**Manifestations** Potential crowding Places of congestion Potential starts of panic Potentially disruptive actions	**Capabilities** Identify and handle crowds Identify and handle congestion Identify and handle panic Identify and handle disruption	**Merit** Value: Locates impediments. Finds and handles crowds, congestion, panic, and disruption Reservation: Requires calibration and clear sight Rebuttal: Many users will accept these limitations in a low-cost system	**Mission** ClusterDetect solves the Problem: - The Manifestations are essential - The Capabilities are efficient and effective - The Merit offered ensures sustained use



Solution Scenarios



Solution Scenarios

Problem/Leverage Scenarios Combined

	Problem (Vertical)	Leverage (Vertical)
Who	Who is affected? Who are the stakeholders?	Who will be part of the contribution? Who are the operators?
Why	Why is this a problem? Is it about money, health, convenience, safety, aesthetics, law, social responsibility, ecology, ...?	Why is this a contribution? Is it bespoke or generic? Is it contextual, universal, preventive, ameliorative, mitigating, compensating, active, passive, supportive, ...?

Vertical Axis

Solution Scenarios

ClusterDetect case

Problem (vertical)

- *Who* has the problem? One option could be an event planner. Another option could be on-the-ground staff handling crowd building.
- *Why* is this a problem? Why are we undertaking this project? The project might focus on a broad problem affecting a market or address the specific challenges of a particular venue.

Leverage (vertical)

- *Who* will operate the system? One option could be having safety assurance specialists work on-site. Another option might be to have on-the-ground staff manage crowd control during an event.
- *Why* is this considered a contribution? Will the system help prevent dangerous situations or assist in managing disasters? Will it focus on a specific scenario or address a broader range of issues? Will it be useful only for disasters, or will it have wider applications?

Vertical Axis

Solution Scenarios 2

	PROBLEM (HORIZONTAL)	LEVERAGE (HORIZONTAL)
What	What type of problem? Is it simple, complicated, complex, or chaotic? (see Table 6.1 on page 84)	What are the inner characteristics of the contribution? Is it flexible, decomposed, integrated, platform-based, open, closed, standardized, ...?
Where	Where do we meet the problem? Where is it located or experienced? Where are those affected by it located?	Where will the contribution be used? Where will it have an effect? What part of the problem is targeted?
When	When does the problem occur? Is it constant, occasional, accidental, random, immediate, delayed, ...?	When will the contribution be deployed? When will the effects be seen (immediately, delayed, or gradually)? Will the contribution be used according to a plan, on demand, constantly, repeatedly, only once, ...?
How	How does the problem cause trouble? Does it affect directly, indirectly, as a network effect, as a side-effect, ...?	How will it work? Will the contribution work directly, indirectly, or via network effects? Will it be active or passive, intrusive or non-intrusive, proactive or reactive, switched or always-on?
How much	How much does the problem bother stakeholders? Can the problem be quantified by what is lost or what could be earned by solving it? Can it be quantified in terms of time, money, manpower, risk, ...? Is it about life, money, or comfort?	How much will the contribution cost? Can the expenses be quantified in terms of investment and operations? Can the contribution be quantified in terms of time, money, manpower, risk, ...?

Horizontal Axis

Solution Scenarios

ClusterDetect case

Problem (horizontal)

- *What* type of problem? This question could concern the project's scope. One option could be *complicated* and limit the scope to addressing only known unknowns. Another option could be *complex* and accept that unknown unknowns might occur. This would allow for a broader scope, greater risk, and, hopefully, a stronger solution.
- *Where* do we meet the problem? Again, this question could affect the project's scope. One option is to focus on *specific venues* to reduce the scope, while another is to aim for *any venue* to increase utility.
- *When* do we meet the problem? The team could consider *predictable situations*, such as breaks or the end of an event. Alternatively, it could involve *accidental situations*, such as a fire breakout or a heart attack.
- *How* does the problem cause trouble? One alternative could be *direct dangers*, such as trampling or stampedes; another could be indirect dangers, such as crowding that prevents paramedics from reaching a visitor experiencing a heart attack.
- *How much* does the problem bother stakeholders? One effect could be *financial losses* if the venue is closed due to safety concerns; another could be *loss of life* in the event of an accident.

Horizontal Axis

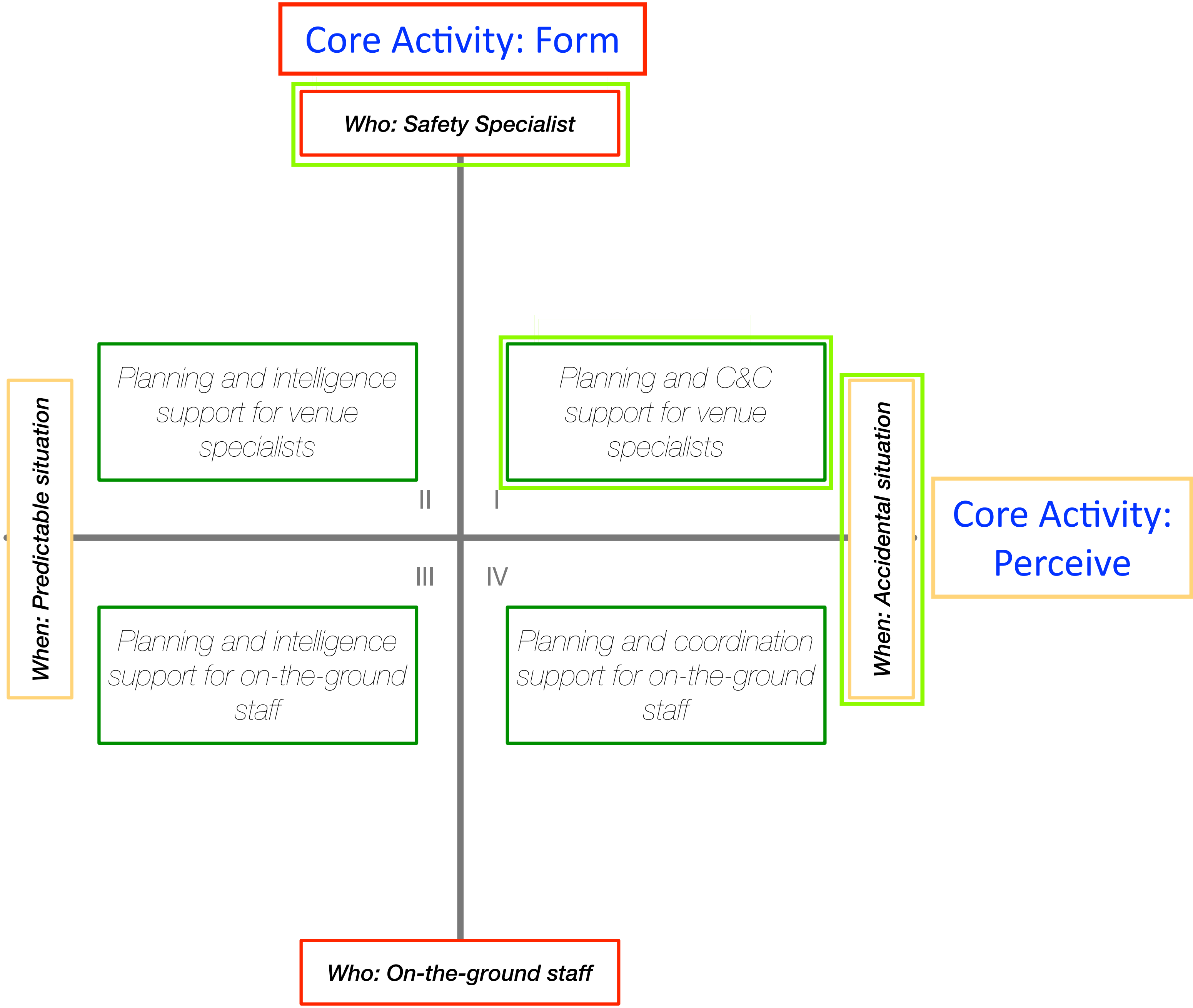
Solution Scenarios ClusterDetect case 2

Leverage (horizontal)

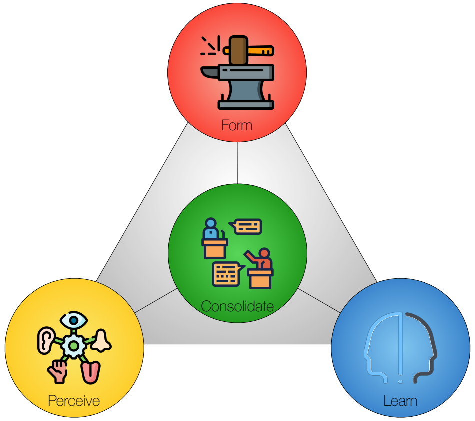
- *What* characterizes the contribution? Is the system highly integrated for efficiency, or is it decomposed for flexibility and a longer lifespan?
- *Where* will the system have an effect? Will it work directly with people, or will it work with the location's infrastructure?
- *When* will the system be used? During planning and prevention work or in critical situations?
- *How* will the system work? Will it passively collect and present information, or will it actively coordinate or control interventions?
- *How much* will the system cost? Will cost considerations mainly focus on investments or operations? Will savings come primarily from reducing manpower costs or from minimizing risks?

Horizontal Axis

Solution Scenarios—ClusterDetect



Example based on one among many alternative axes



Next Time

Next Exercise and Lecture

- Exercise 5:
  - Solution Scenarios.
- Lecture 6:
  - Chapter 7 in Essence: *Horizon*.

Exercise 5 (Solution Scenarios)

Use the results from Exercise 4 in this exercise.

- Develop axes for the Solution Scenarios as described in Section 8.3 (Tables 8.2 and 8.3).
- Develop Solution Scenarios as illustrated in Figure 8.1.
- Explain the Solution Scenarios you find in each quadrant.
- Choose one scenario to work with from now on. You can choose one quadrant or combine several if relevant.
- Discuss short-term strengths and weaknesses of your chosen scenario–how might this scenario create value and growth in the short run?
- Discuss longer-term opportunities and threats related to your chosen scenario–how might the future look if you use this scenario?